

CS & IT ENGINEERING



Computer Network - 2

Framing & Error Control

Lecture No. - 01

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Recap of Previous Lecture



Topic

Ethernet Standard

Topic

Ethernet Bridging



Topics to be Covered



Topic

Framing





Topic : LAN Standards



- IEEE 802.3 : Ethernet [Bus Topology => CSMA/CD]
- IEEE 802.4 : Token Bus [Bus Topology => Token]
- IEEE 802.5 : Token Ring [Ring Topology => Token]
- IEEE 802.11 : Wireless [Wi-Fi => CSMA/CA]

[MCQ]

[GATE-IT-2005] [1 Mark]



#Q. Which of the following statements is TRUE about CSMA/CD?

- A IEEE 802.11 wireless LAN runs CSMA/CD protocol. F [CSMA/CA]
- B Ethernet is not based on CSMA/CD protocol. F
- ✓ C CSMA/CD is not suitable for a high propagation delay network like satellite network. T $\Rightarrow$ ALOHA
- D There is no contention in a CSMA/CD network. F

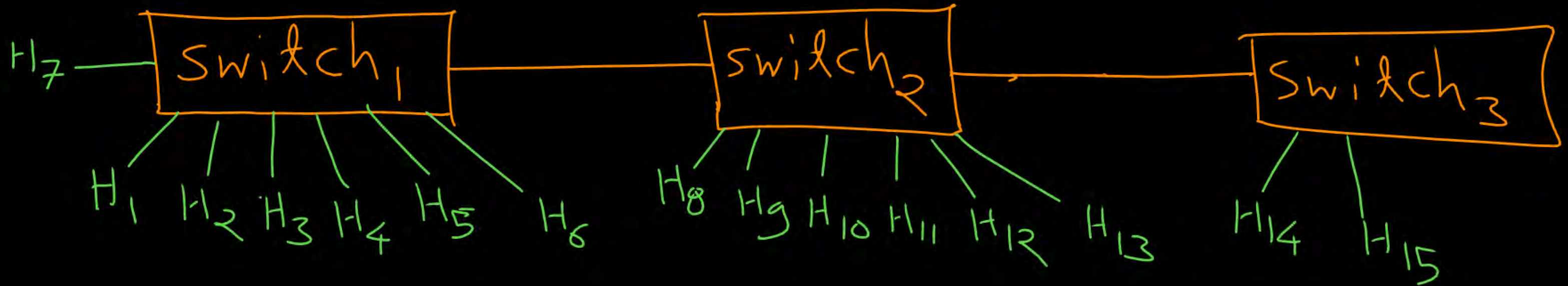
collision slot $\Rightarrow$ contention

[NAT]

[GATE-2019][2 Mark]



#Q. Consider that 15 machines need to be connected in a LAN using 8-port Ethernet switches. Assume that these switches do not have any separate uplink ports. The minimum number of switches needed is _____.



Ans = 3



Topic : Hub

⇒ No store-and-forward



- Not a broadcast domain separator
[Entire LAN has same broadcast domain]
- Not a collision domain separator
[Entire LAN has same collision domain]

Collision domain : Place where collision can occur



Topic : Switch

Bridge $\Rightarrow$ Store-and-Forward



- Not a broadcast domain separator
[Entire LAN has same broadcast domain]
- Collision domain separator
[Different LAN segment has different collision domain]

Collision domain : Place where collision can occur

* Transparent Switch



Topic : Router

⇒ Store-and-Forward



→ Broadcast domain separator

[Different subnets have different broadcast domain]

→ Collision domain separator

[Different subnets have different collision domain]

Collision domain : Place where collision can occur

[MCQ]

[GATE-IT-2005] [1 Mark]



#Q. Which of the following statements is FALSE regarding a bridge?

- A Bridge is a layer 2 device T
- B Bridge reduces collision domain T
- C Bridge is used to connect two or more LAN segments T
- ✓ D Bridge reduces broadcast domain F

Ans: D

#Q. Ethernet layer-2 switch is a network element type which gives

- ☒ A different collision domain and same broadcast domain
- ☐ B different collision domain and different broadcast domain
- ☐ C same collision domain and same broadcast domain
- ☐ D same collision domain and different broadcast domain

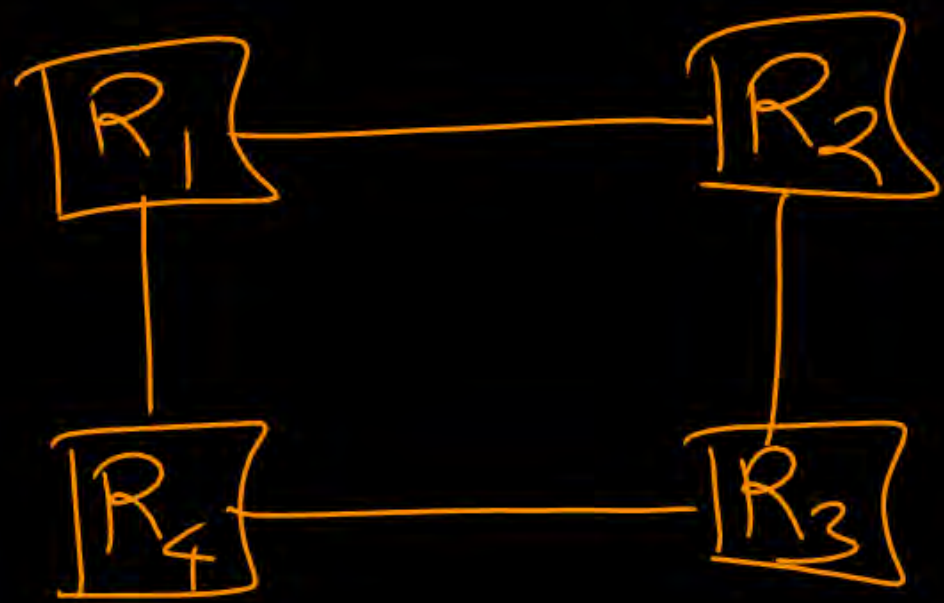
Ans: A



Topic : STP



- Multiple switches inside a network may form circuit (loop)
- To avoid indefinite traversing of frames
switches uses "Spanning Tree Protocol (STP)"



→ To prevent
IP packet
indefinite traversing

IP uses TTL (Time-to-Live)
(Hop-Limit)



→ To prevent
indefinite traversing
of frame,

Switches uses STP.

[MCQ]

[GATE-CS-2005] [1 Mark]



#Q. In a network of LANs connected by bridges, packets are sent from one LAN to another through intermediate bridges. Since more than one path may exist between two LANs, packets may have to be ruoted through multiple bridges. Why is the spanning tree algorithm used for bridge-routing?

- ☒ A For shortest path routing between LANs
- ☒ B For avoiding loops in the routing paths
- ☒ C For fault tolerance
- ☒ D For minimizing collisions

Ans: B



Topic : Syllabus



Concept of layering : OSI and TCP/IP Protocol Stacks;

Basics of packet, circuit and virtual circuit switching;

Data link layer : framing, error detection, Medium Access Control,
Ethernet bridging;



Topic : Framing

⇒ Frame Synchronization



Problem : How receiver identify frame boundaries while receiving multiple frames?

[Variable length frames and transmitted without time-gap]

↓
No any interframe gap.



Topic : Framing

⇒ Point-to-Point communication





Topic : Framing



Problem : How receiver identify **frame boundaries** while receiving **multiples frames**?

[Variable length frames and transmitted without time-gap]

Solution :

1. Byte (Character) Count

2. Byte (Character) Stuffing

3. Bit Stuffing

CS-04, CS-14

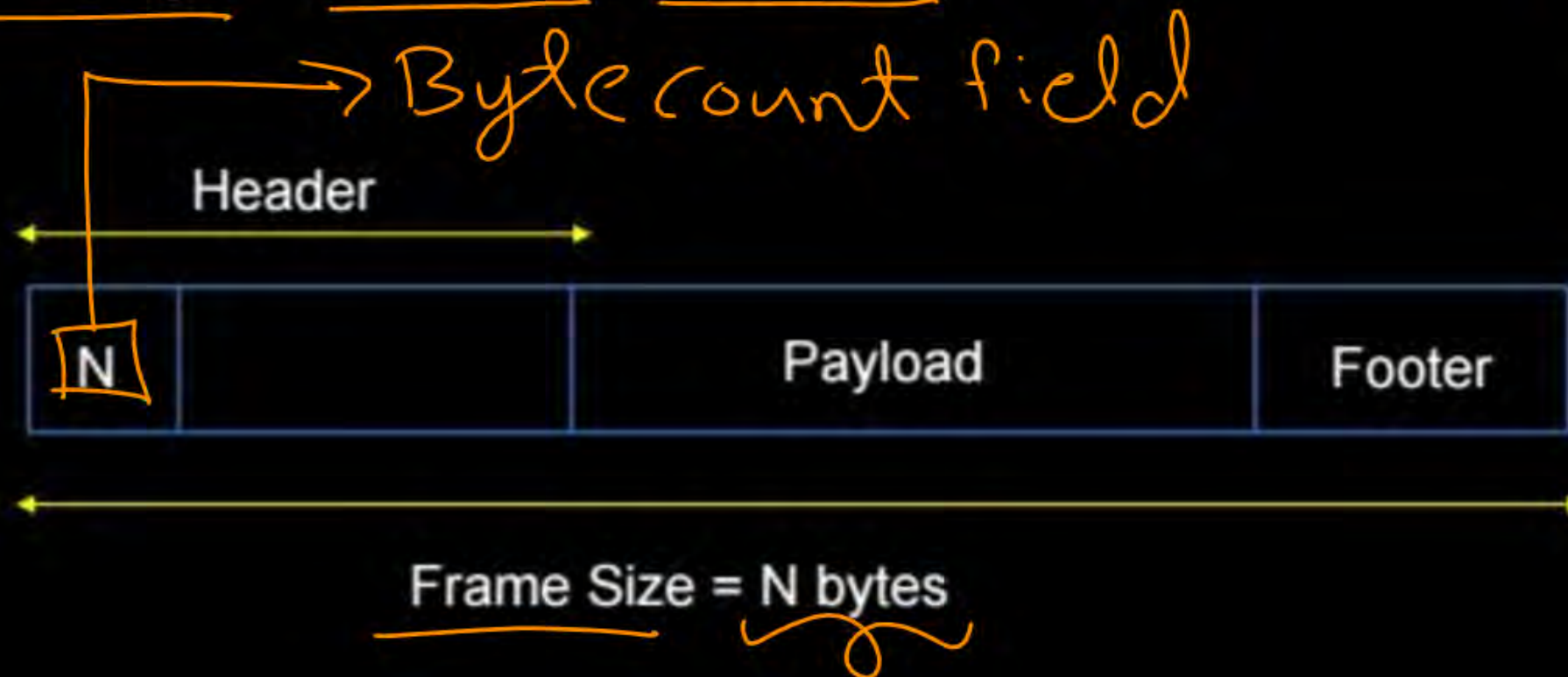


Topic : Byte Count



Byte Count field :

- One or two byte in size
- At the begin of the frame [Part of Header]
- Contains length of frame in bytes
[Size of Byte Count field inclusive]



16 bit Range

$0 \text{ to } (2^{16} - 1)$

→ $(2^{16} - 1)$ byte

8-bit Range

$0 \text{ to } (2^8 - 1)$

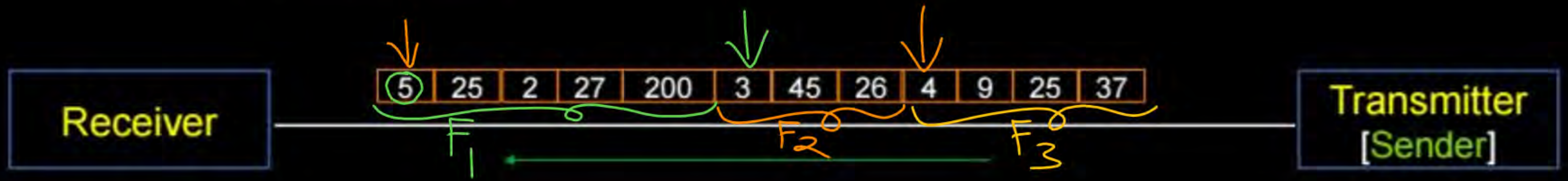
→ 255 byte



Topic : Byte Count

char count

=> 1 byte



5 → 101
4 → 100

Ethernet Frame
↳ Type



Topic : Byte Count



Disadvantage :

if error occurred in the “Byte Count field”

then entire transmission is garbled [that frame onwards]

[Need to retransmit more than one frame]



Topic : Byte Stuffing



Special characters :

1. **"STX"** : [Start of Text / Transmission]
Start Frame Delimiter (SFD), ASCII value = 2
2. **"ETX"** : [End of Text / Transmission]
End Frame Delimiter (EFD), ASCII value = 3
3. **"ESC"** : [Escape character]
ASCII value = 27



Topic : Byte Stuffing

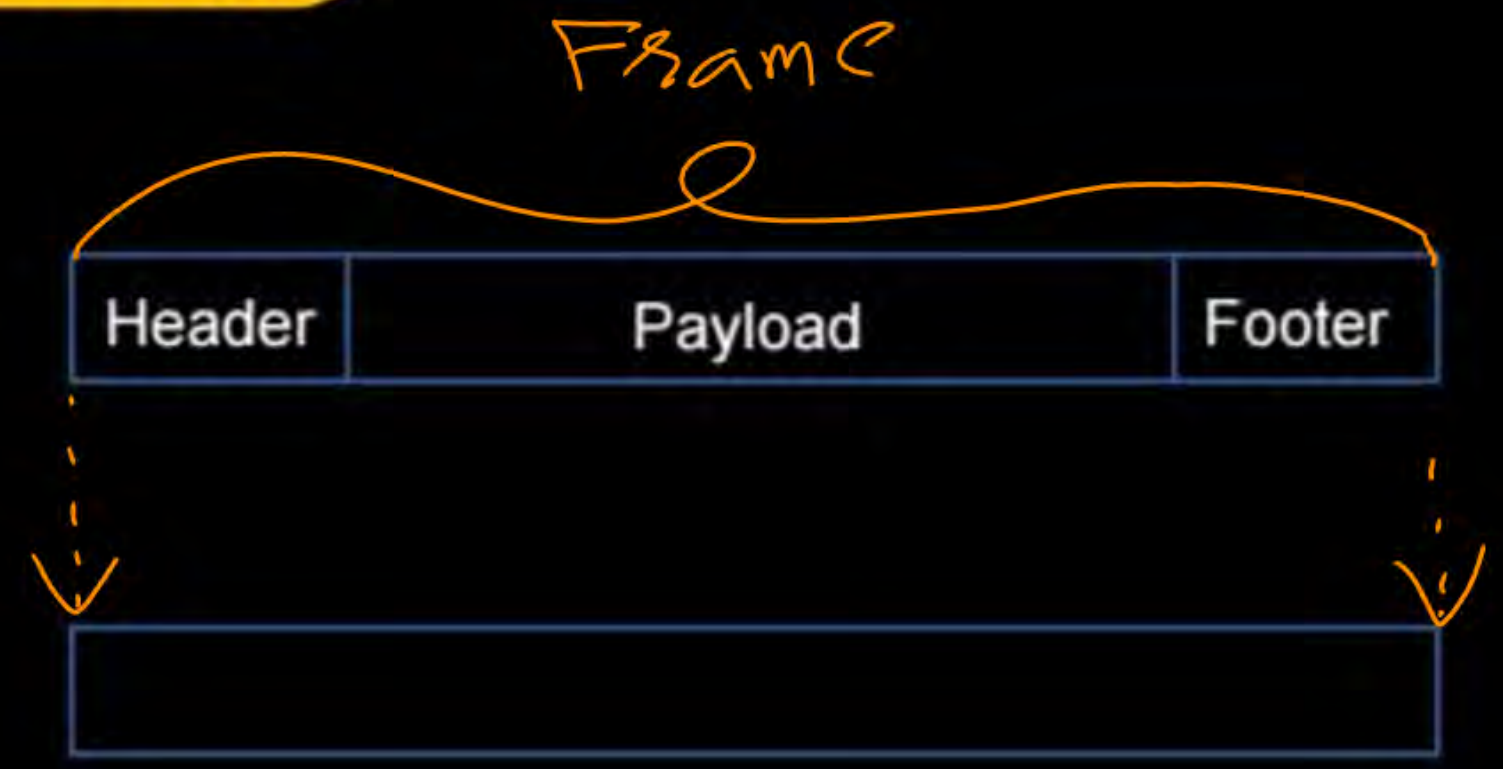


Transmitter Protocol :

1. Transmit **“STX”** just before frame transmission start.
2. Transmit **“ETX”** just after frame transmission completed.
3. Stuff **“ESC”** character just before every **“STX”** / **“ETX”** / **“ESC”** character appear in data while transmission. [Except start and end of frame]

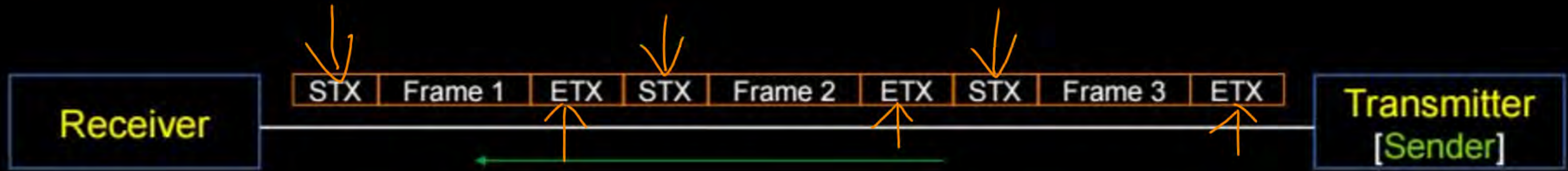


Topic : Byte Stuffing





Topic : Byte Stuffing





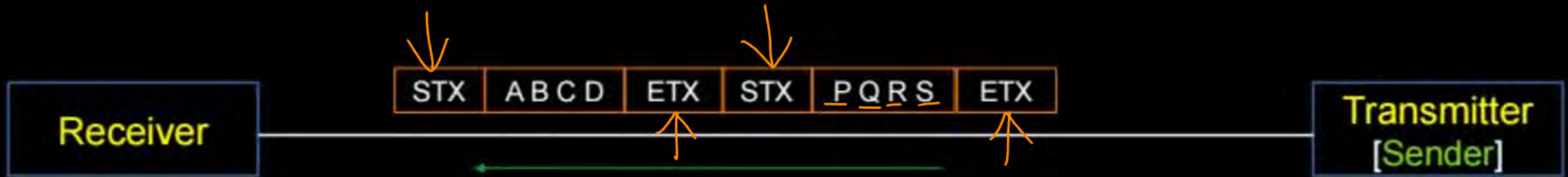
Topic : Byte Stuffing



CASE I :

Frame 1 : "A B C D"

Frame 2 : "P Q R S"





Topic : Byte Stuffing

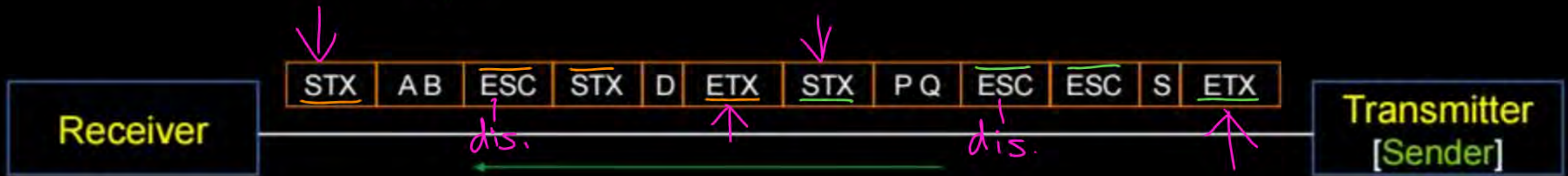
(char stuffing)



CASE II :

Frame 1 : "A B STX D"

Frame 2 : "P Q ESC S"



$F_1 = A B S T X D$

$F_2 = P Q E S C S$

Frame = AB ESC ETX

STX AB ESC ESC ESC ETX ETX



Topic : Byte Stuffing



Receiver Protocol :

1. Looking for **"STX"** (start of frame)
2. After **"STX"**, looking for **"ETX"** (End of frame)
3. Discard **"ESC"** character while receiving the data frame and copy next character into the receiving buffer



Topic : Byte Stuffing



Disadvantage :

Every "STX" / "ETX" / "ESC" (special character) in the data frame will increase length of frame by "one-byte".

[Except Start and End of Frame]



2 mins Summary



Topic

Byte Count ✓

Topic

Byte Stuffing ✓

Topic

~~Bit Stuffing~~



THANK - YOU

